

Mathematical and Computational Methods

Exercise Sheet 10: Integration II: techniques and applications

Tutorial route (►, in this order, about 80 minutes): 1.2, 2.1(a,b), 2.5(a), 2.4(a), 2.8(a,c), 2.9(a,b), 2.10(a). Everything else is for home study.

Levels: ★ drill, ★★ core, ★★ challenge. Give exact answers unless a decimal is asked for.

1 Warm-up drills

Find the indefinite integral (with +C) unless limits are shown.

1.1 Techniques

Exercise 1.1 (★ drill — by parts). (a) $\int x e^{2x} dx$, (b) $\int x \ln x dx$.

Exercise 1.2 (► ★ drill — partial fractions). $\int \frac{3x+1}{x(x+1)} dx$.

Exercise 1.3 (★ drill — trig integrals). (a) $\int \sin^2 x dx$, (b) $\int \cos^3 x dx$, (c) $\int \tan^2 x dx$.

Exercise 1.4 (★ drill — trig substitution). $\int \frac{dx}{\sqrt{9-x^2}}$.

1.2 Applications

Exercise 1.5 (★ drill — area between curves). Find the area enclosed between $y = 2x$ and $y = x^2$.

Exercise 1.6 (★ drill — volume of revolution). The region under $y = \sqrt{x}$, $0 \leq x \leq 4$, is revolved about the x -axis. Find the volume.

Exercise 1.7 (★ drill — average value). Find the average value of $f(x) = x^2$ on $[0, 3]$.

Exercise 1.8 (★ drill — density). The random variable X has density $f(x) = 3x^2$ on $[0, 1]$ (and 0 elsewhere). Verify it is a density, then find $P(X > \frac{1}{2})$ and $\mathbb{E}[X]$.

2 Tutorial problems

Exercise 2.1 (► ★★ core — by parts (worked-example anchor)). Evaluate

$$(a) \int x^2 e^x dx, \quad (b) \int \arctan x dx, \quad (c) \int e^x \sin x dx.$$

Exercise 2.2 (★★ core — partial fractions). Evaluate

$$(a) \int \frac{2x+3}{x(x+1)^2} dx, \quad (b) \int \frac{dx}{x^2+4x+13}, \quad (c) \int \frac{x^3}{x^2+1} dx.$$

Exercise 2.3 (★★core — *trigonometric integrals*). Evaluate

$$(a) \int_0^{\pi/2} \sin^3 x \cos^2 x \, dx, \quad (b) \int_0^{\pi/4} \tan^3 x \, dx, \quad (c) \int \sec^4 x \, dx.$$

Exercise 2.4 (► ★★core — *trigonometric substitution*). Evaluate

$$(a) \int \sqrt{9 - x^2} \, dx, \quad (b) \int \frac{dx}{(x^2 + 4)^{3/2}}.$$

Exercise 2.5 (► ★★core — *area between curves (worked-example anchor)*).

- Sketch the region first, then find the area enclosed between the parabola $y = x^2$ and the line $y = x + 2$.
- Find the total area between $y = \sin x$ and $y = \cos x$ for $0 \leq x \leq \frac{\pi}{2}$.

Exercise 2.6 (★★core — *volumes of revolution*).

- Find the volume when the region under $y = \sin x$, $0 \leq x \leq \pi$, is revolved about the x -axis.
- Revolving $y = \sqrt{r^2 - x^2}$, $-r \leq x \leq r$, about the x -axis sweeps out a sphere of radius r . The cap of height h ($0 < h \leq 2r$) is the part with $r - h \leq x \leq r$. Show that its volume is $V = \frac{1}{3}\pi h^2(3r - h)$, and check that $h = 2r$ gives the whole sphere.
- Spot the error. For the region of Warm-up 1.6 a student writes $V = \pi \int_0^4 \sqrt{x} \, dx = \pi \left[\frac{2}{3}x^{3/2} \right]_0^4 = \frac{16\pi}{3}$, instead of the 8π found there. What went wrong, and what is the correct set-up?

Exercise 2.7 (★★core — *improper integrals*). Evaluate, or show divergent:

$$(a) \int_0^\infty e^{-2x} \, dx, \quad (b) \int_0^\infty x e^{-x^2} \, dx, \quad (c) \int_1^\infty \frac{dx}{x^p} \quad (p > 0).$$

Exercise 2.8 (► ★★core — *probability and expectation (data science)*). (Exponential law.) The lifetime T (hours) of a component has density $f(t) = \lambda e^{-\lambda t}$ on $[0, \infty)$ with $\lambda = \frac{1}{2}$.

- Find $P(T > 2)$.
- Find $P(1 < T < 3)$.
- Find the mean lifetime $\mathbb{E}[T]$.

Exercise 2.9 (► ★★core — *setting up (physics, work)*). A uniform rope of length $L = 10$ m and mass $\mu = 2$ kg per metre hangs from a winch at the top of a building. Take $g = 9.8 \text{ m s}^{-2}$.

- Find the work done to wind the whole rope to the top.
- Find the average depth $\bar{x}$ below the top of the points of the hanging rope, and show that the work in (a) equals the rope's weight $\mu g L$ times $\bar{x}$.

Exercise 2.10 (► ★★core — *volume by cross-sections*).

- A pyramid of height h has as its base an equilateral triangle of side a . Measuring x down from the apex, use similar triangles to find the area $A(x)$ of the cross-section at depth x , and show that $V = \frac{\sqrt{3}}{12}a^2h$.
- ★★★challenge A solid has a circular base of radius R in the xy -plane. Every cross-section perpendicular to a fixed diameter is a square. Find its volume.

3 Further practice (home)

3.1 Techniques

Exercise 3.1 (★★ core — linear over quadratic). $\int \frac{2x+3}{x^2+9} dx$.

Exercise 3.2 (★★ core — trig substitution (secant)). $\int \frac{dx}{\sqrt{x^2-9}}$ (for $x > 3$).

3.2 Applications

Exercise 3.3 (★★ core — average value (physics)). A particle moves with velocity $v(t) = 6t - t^2$ (m s^{-1}) for $0 \leq t \leq 6$. Find its average velocity over this interval.

Exercise 3.4 (★★ core — find the constant). Find k so that $f(x) = kx$ is a probability density on $[0, 4]$, then compute $\mathbb{E}[X]$.

Exercise 3.5 (★ drill — setting up).

- A rod occupies $0 \leq x \leq L$ with linear density $\rho(x) = kx$. Find its mass.
- (Physics.) A spring of stiffness 200 N m^{-1} needs the force $F(x) = 200x$ to hold it stretched by x metres. Find the work done in stretching it from its natural length to $x = 0.1 \text{ m}$.

3.3 Conceptual

Exercise 3.6 (★★ core — true or false?). Decide, with a one-line reason, whether each statement is true or false.

- Every rational function has an elementary antiderivative (after polynomial division, if needed).
- $\int_{-\infty}^{\infty} x \, dx = 0$ because the integrand is odd.
- The average value of a continuous f on $[a, b]$ always lies between the minimum and maximum of f on $[a, b]$.

3.4 Challenges and extensions

Exercise 3.7 (★★★ challenge — mass and centre of mass). A rod on $0 \leq x \leq L$ has density $\rho(x) = \rho_0(1 + \frac{x^2}{L^2})$. Find its mass M and the position $\bar{x} = \frac{1}{M} \int_0^L x \rho(x) \, dx$ of its centre of mass.

Exercise 3.8 (★★★ challenge — normalising the Gaussian (data science)). Given the standard result $\int_{-\infty}^{\infty} e^{-x^2} \, dx = \sqrt{\pi}$, find the constant c making $f(x) = c e^{-x^2}$ a probability density on $\mathbb{R}$, and find $\mathbb{E}[X]$.